

Decent Coaching Classes

X SB

Mind Your Math - VII

Answer Key & Marking Scheme

Marks: 40

Time: 2 Hrs

Q.1 (A) Correct alternatives [4]

- (1) (A) -14 ($D = 2 \times (-1) - 3 \times 4 = -14$) [1]
- (2) (A) real and equal ($b^2 - 4ac = 16 - 16 = 0$) [1]
- (3) (B) 46 ($4 + 14 \times 3$) [1]
- (4) (A) $1/2$ (primes $2, 3, 5 \rightarrow 3/6$) [1]

Q.1 (B) Solutions [4]

- (1) $S_n = n/2 (a + t_n)$ [1]
- (2) $S = \{HH, HT, TH, TT\}$ [1]
- (3) $NAV = 300/10 = ₹30$ per unit [1]
- (4) Class mark $= (150 + 200)/2 = 175$ [1]

Q.2 (A) Any two (2 marks each) [4]

- (1) $D = -2 - 3 = -5$; $D_x = -6 - 4 = -10$; $D_y = 4 - 9 = -5$ [1]
 $x = -10/-5 = 2$; $y = -5/-5 = 1$ [1]
- (2) $m^2 - 13m - m + 13 = 0$ [$1/2$]
 $m(m - 13) - 1(m - 13) = 0$ [$1/2$]
 $(m - 13)(m - 1) = 0$, $m = 13$ or $m = 1$ [1]
- (3) $d = 4$ [$1/2$]
 $S_{20} = 10 [10 + 19 \times 4] = 10 \times 86$ [1]
 $= 860$ [$1/2$]

Q.2 (B) Any two (2 marks each) [4]

- (1) Multiply second equation by 3: $3x - 3y = 3$. Adding to $2x + 3y = 12$ gives $5x = 15$ [1]
 $x = 3$; then $y = x - 1 = 2$. $(x, y) = (3, 2)$ [1]
- (2) $a = 2, b = 4, c = 5$. $b^2 - 4ac = 16 - 40 = -24$ [1]
Discriminant is negative, so the roots are not real. [1]
- (3) $n(S) = 52$. (i) $n(A) = 4, P(\text{king}) = 4/52 = 1/13$ [1]
(ii) $n(B) = 26, P(\text{red card}) = 26/52 = 1/2$ [1]

Q.3 (A) Any one (3 marks) [3]

- (1) CGST rate = 6%, SGST rate = 6% [1]
CGST = ₹300, SGST = ₹300, total GST = ₹600 [1]
Invoice value $= 5000 + 600 = ₹5600$ [1]
- (2) $x_i = 5, 15, 25, 35, 45$; $\Sigma f_i = 50$ [1]
 $\Sigma f_i x_i = 25 + 120 + 375 + 560 + 270 = 1350$ [1]
Mean $= 1350/50 = 27$ [1]

Q.3 (B) Any two (3 marks each) [6]

- (1) $a = 2, b = -7, c = 3$; $b^2 - 4ac = 49 - 24 = 25$ [1]
 $x = (7 \pm 5)/4$ [1]

$$x = 3 \text{ or } x = 1/2 \quad [1]$$

- (2) $a = 7, d = 6, t_n = 205. \quad t_n = a + (n - 1)d \quad [1]$
 $205 = 7 + (n - 1) \times 6, \text{ so } 198 = 6(n - 1) \quad [1]$
 $n - 1 = 33, n = 34. \text{ There are 34 terms.} \quad [1]$

- (3) $n(S) = 36 \quad [1]$
Sum 9: (3,6), (4,5), (5,4), (6,3), so $n(A) = 4 \quad [1]$
 $P(A) = 4/36 = 1/9 \quad [1]$

Q.4 Any three (4 marks each)

[12]

- (1) Let ten's digit = x , unit's digit = $y. \quad x + y = 9 \quad [1]$
 $(10y + x) - (10x + y) = 27, \text{ so } y - x = 3 \quad [1]$
Adding: $2y = 12, y = 6, x = 3 \quad [1]$
Original number = 36 $[1]$
- (2) Let the numbers be x and $x + 1. \quad x^2 + (x + 1)^2 = 113 \quad [1]$
 $2x^2 + 2x - 112 = 0, \text{ so } x^2 + x - 56 = 0 \quad [1]$
 $(x + 8)(x - 7) = 0, x = -8 \text{ (rejected) or } x = 7 \quad [1]$
The numbers are 7 and 8. $[1]$
- (3) Output GST = 12% of 70,000 = ₹8400 $[1]$
Input tax credit (ITC) = 12% of 50,000 = ₹6000 $[1]$
GST payable = 8400 - 6000 = ₹2400 $[1]$
CGST payable = SGST payable = ₹1200 each $[1]$
- (4) Class marks $x_i = 15, 25, 35, 45, 55. \text{ Take } A = 35; d_i = -20, -10, 0, 10, 20 \quad [1]$
 $f_i d_i = -80, -100, 0, 80, 40; \Sigma f_i = 40, \Sigma f_i d_i = -60 \quad [1]$
Mean of deviations = $-60/40 = -1.5 \quad [1]$
Mean = $A + \text{mean of deviations} = 35 - 1.5 = 33.5 \quad [1]$
- (5) A.P.: $a = 1000, d = 200, n = 12 \quad [1]$
 $S_{12} = 12/2 [2 \times 1000 + 11 \times 200] \quad [1]$
 $= 6 \times 4200 \quad [1]$
 $= ₹25,200 \quad [1]$

Q.5 Any one (3 marks)

[3]

- (1) (i) Brokerage per share = 0.5% of 150 = ₹0.75; cost per share = ₹150.75 $[1]$
Total amount = $40 \times 150.75 = ₹6030 \quad [1]$
(ii) Dividend = 12% of 100 = ₹12 per share; $40 \times 12 = ₹480 \quad [1]$
- (2) Units = $6000 / 24 = 250 \text{ units} \quad [1]$
Present value = $250 \times 28.50 = ₹7125 \quad [1]$
Gain = $7125 - 6000 = ₹1125 \quad [1]$